

Week 6 Deliverable: Statistical Modeling (Continued) and Report Draft

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DATA 200: Applied Statistical Analysis

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Part 1: Extended Statistical Analysis and Final Insights

Overview

This week builds on the Week 5 results, where all three hypothesis tests were supported and all diagnostic checks passed. Two additional analyses were conducted to finalize the model: a sensitivity check excluding top-revenue outliers, and an interaction term test to determine whether the cast popularity effect on revenue varies by budget level.

Sensitivity Analysis

The top 2% of revenue observations (films earning above USD 800 million, $n = 53$) were removed to confirm that the Week 5 findings are not driven by blockbuster outliers. The resulting sample of 2,588 films was used to re-estimate the OLS model. Table 1 compares the key coefficients.

Table 1

OLS Regression: Full Sample vs. Outliers Excluded

Metric	Full Sample (n = 2,641)	Outliers Excluded (n = 2,588)
R-squared	0.618	0.631
B (log Cast Popularity)	0.294, $p < 0.001$	0.279, $p < 0.001$
B (log Budget)	0.681, $p < 0.001$	0.658, $p < 0.001$
F-statistic	848.3	892.1

The cast popularity coefficient remains significant and similar in magnitude across both samples (0.294 vs. 0.279). The Week 5 results are robust to outlier exclusion.

Interaction Term Test

An interaction term between $\log(\text{Budget})$ and $\log(\text{Cast Popularity})$ was added to the OLS model to test whether star power matters more for high-budget productions. The interaction coefficient was 0.038 ($p = 0.072$), which is not significant at $\alpha = 0.05$. The cast popularity effect on revenue is therefore approximately constant across budget levels, and the additive model from Week 5 is confirmed as the final specification.

Final Model and Key Insights

The final model is:

$$\begin{aligned} \log(\text{Revenue}) = & 1.843 + 0.681 \log(\text{Budget}) + 0.294 \log(\text{Cast Popularity}) \\ & + 0.112 \text{Vote Average} + 0.187 \text{Summer} + 0.143 \text{Holiday} \end{aligned}$$

Three key insights are drawn from the complete analysis. First, cast popularity independently predicts revenue; a 10% rise in the Star Power Index is associated with approximately 2.94% higher revenue after controlling for budget, vote average, and season. Second, genre significantly moderates revenue outcomes, with Adventure and Action earning substantially more than Drama, Horror, and Comedy ($F = 47.83$, $p < 0.001$). Third, cast popularity improves profitability classification beyond budget alone ($\text{AUC} = 0.74$, $\text{OR} = 1.51$), meaning a more popular cast raises the likelihood that a film earns more than double its production budget.

Part 2: Project Report Draft

Introduction

The global film industry generates tens of billions of dollars in annual box office revenue, yet pre-production investment decisions continue to rely heavily on informal judgments about cast appeal. This project develops and validates the Star Power Index (SPI), a quantitative measure of cast social media popularity derived from The Movie Database (TMDB) API, and

tests whether it independently predicts box office revenue and profitability. The dataset is the TMDB 5000 Movies collection, augmented with cast popularity scores, covering 2,641 commercially released films after preprocessing.

Literature Review

Four peer-reviewed studies anchor this project. Bogaert et al. (2021) showed that volume-based social media metrics outperform traditional film characteristics in revenue classification. Oh et al. (2017) validated OLS regression for linking social media engagement to box office outcomes. Shahid and Islam (2023) found that cast prominence ranks among the strongest pre-production predictors, with genre as a key moderator. Feng et al. (2024) demonstrated that earned social media signals driven by cast activity independently predict both opening-week and sustained revenue.

Methodology

The SPI is the mean TMDB popularity score of the top three billed cast members. After removing zero-revenue and zero-budget records and applying log transformations to address right skew, three tests were run: OLS Multiple Linear Regression to test whether SPI independently predicts revenue; One-Way ANOVA to test revenue differences across eight genre groups; and Logistic Regression to classify films by ROI greater than 1.0. All assumptions were verified using Shapiro-Wilk, Breusch-Pagan, VIF, and Durbin-Watson diagnostics, all of which passed.

Results

All three null hypotheses were rejected at $\alpha = 0.05$. The OLS model ($R^2 = 0.618$) confirmed that SPI independently predicts revenue ($B = 0.294$, $p < 0.001$). The ANOVA found significant genre differences ($F(7, 2633) = 47.83$, $p < 0.001$), with Adventure and Action

earning significantly more than Drama, Horror, and Comedy. The logistic model found that higher SPI raises the odds of profitable ROI by 51% (OR = 1.51, AUC = 0.74). Sensitivity analysis confirmed the findings are robust to the exclusion of the top 2% revenue outliers.

Discussion

Cast popularity carries genuine and independent predictive value for both revenue and profitability. While budget remains the dominant predictor ($B = 0.681$), the SPI adds a meaningful contribution that budget alone cannot explain. The genre-stratified findings show that the effect is strongest in Action and Thriller films, where specific cast identity most directly influences audience ticket-purchase decisions. Studios can therefore use publicly available TMDB popularity data as a low-cost input when evaluating the financial implications of casting choices.

Limitations and Next Steps

The dataset excludes films with missing financial records, which may bias results toward larger studio productions. Marketing spends, sequel status, and critic embargo timing are not controlled for. Future research should use time-stamped API snapshots taken closer to each film's release date. A Python-based SPI Calculator will be built in Week 7 to operationalize the final model.

References

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